

Inverse & Composite Functions

ANSWER KEY



Composing functions

- Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.
 $(f \circ g)(x) = f(g(x)) = 2(x^2 - 1) + 3 = 2x^2 + 1$.
 $(g \circ f)(x) = g(f(x)) = (2x + 3)^2 - 1 = 4x^2 + 12x + 9 - 1 = 4x^2 + 12x + 8$.
- Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.
 $(f \circ g)(2) = f(g(2)) = f(0) = \sqrt{0 + 4} = 2$. In general, $f \circ g$ requires $g(x) + 4 \geq 0$, i.e. $x^2 \geq 0$, which holds for all real x — so $f \circ g$ is defined everywhere.

Decomposing a composite function

- Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g .
Let $g(x) = 3x - 5$ (inner function) and $f(x) = x^4$ (outer function). Then $f(g(x)) = (3x - 5)^4 = h(x)$.
- Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.
Let $g(x) = x^2 + 1$ and $f(x) = \sqrt{x}$. Then $f(g(x)) = \sqrt{x^2 + 1} = h(x)$.

Verifying inverse functions

- Verify algebraically that $f(x) = \frac{x-2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.
 $f(g(x)) = \frac{(3x+2)-2}{3} = \frac{3x}{3} = x$. $g(f(x)) = 3\left(\frac{x-2}{3}\right) + 2 = (x-2) + 2 = x$. Since both compositions return x , f and g are inverses.
- Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.
Solve $y = 2x^3 + 1$ for x : $x^3 = \frac{y-1}{2}$, so $f^{-1}(x) = \sqrt[3]{\frac{x-1}{2}}$. Check at $x = 17$: $f^{-1}(17) = \sqrt[3]{\frac{17-1}{2}} = \sqrt[3]{8} = 2$, and $f(2) = 2(8) + 1 = 17$ ✓.

Visualizing inverse functions as reflections

- 7 The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.
- a** State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why. $f(x) = x^2$ must be restricted to $x \geq 0$ (or $x \leq 0$) because the full parabola fails the horizontal line test — without restricting the domain, f is not one-to-one and has no inverse function.
- b** Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically. Setting $f(x) = x$: $x^2 = x$, so $x^2 - x = 0$, $x(x - 1) = 0$, giving $x = 0$ or $x = 1$. Both points $(0, 0)$ and $(1, 1)$ lie on $y = x$, matching where the two curves cross in the graph.

Domain and range of composite and inverse functions

- 8 Given $f(x) = \frac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.
 $(f \circ g)(x) = f(x + 1) = \frac{1}{(x + 1) - 3} = \frac{1}{x - 2}$. This is undefined at $x = 2$, so the domain is all reals except $x = 2$.
- 9 If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1} .
 The domain and range swap for an inverse function: f^{-1} has domain $[0, \infty)$ and range $[1, \infty)$.

Applying composite and inverse functions in context

- 10 A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.
 Solve $C = \frac{5}{9}(F - 32)$ for F : $\frac{9C}{5} = F - 32$, so $F = \frac{9}{5}C + 32$. This is C^{-1} , which converts a Celsius temperature back to Fahrenheit.

Using the horizontal line test

- 11 Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.
 Since $f(x) = x^3 - 4x$ has multiple turning points (a cubic with two real critical points), a horizontal line such as $y = 0$ intersects the graph at more than one point ($x = 0, \pm 2$), failing the horizontal line test. So f does not have an inverse function over its entire domain.
- 12 Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x , and find its inverse.
 $f(x) = 5x - 2$ is a line with nonzero slope, so it is strictly increasing and every horizontal line crosses it exactly once — it passes the horizontal line test everywhere. Solving $y = 5x - 2$ for x : $f^{-1}(x) = \frac{x + 2}{5}$.

Composite functions with restricted domains

- 13** Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

$(f \circ g)(x) = f(x - 4) = \frac{1}{x - 4}$. This is undefined at $x = 4$, so the domain is all reals except $x = 4$.